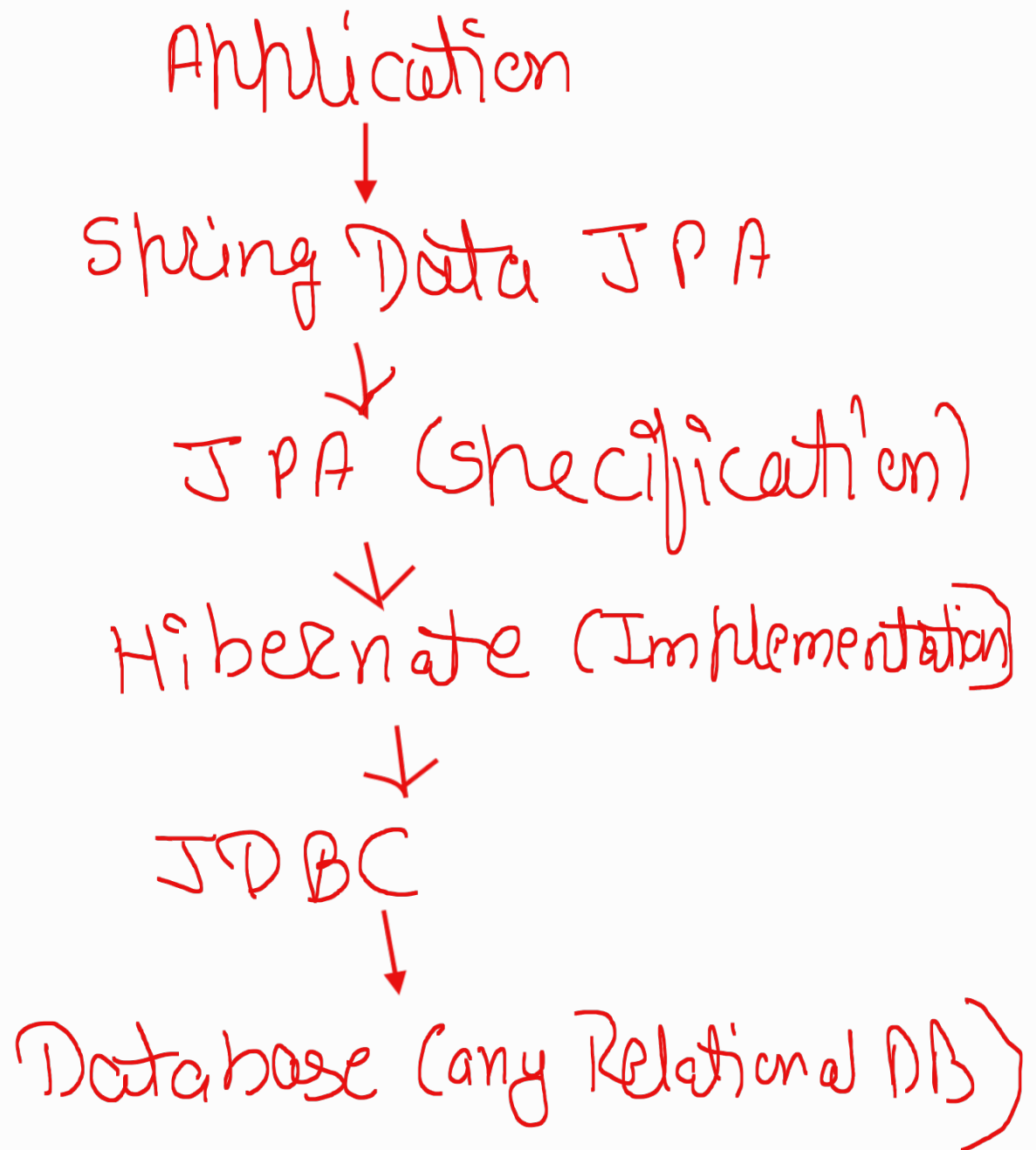


Relation betⁿ these three



① JPA (Java Persistence API)

→ It is a specification
(set of rules & interfaces)
that defines how objs should be
mapped to relational DB

→ JPA is like a blueprint

It tell developers

- How to map Java Classes to DB tables
- which annotations to use
- which method should be available
- How persistence should work

→ JPA cannot work alone it needs an implementation such as Hibernate.

(2) Hibernate

→ It is an ORM framework & most popular framework of JPA

→ Hibernate follows all JPA rules & provides the actual code that performs DB operations

→ Features of Hibernate are:

- Automatic SQL Generation
- Lazy Loading
- Eager Loading
- First level cache
- Second level cache
- Transaction management
- HQL
- Native sql support.

Key Points:

- It implements JPA
- It is an ORM framework
- Can also be used with JPA using Hibernate-specific APIs (like Session), although using JPA APIs is modern, portable approach

③ Spring Data JPA

- Spring data JPA is a spring module built on top of JPA
- It further reduces the amt of code developers need to write
 - Without Spring Data JPA, you would manually use EntityManager
 - With Spring Data JPA, you simply create an interface

Without SDJ,

```
Student s = entityManager.find(Student.class, 1);
```

with SDJ,

```
public interface StudentRepo extends  
    JPARepository<Student,  
        Long> {
```

}

Now, you can do,

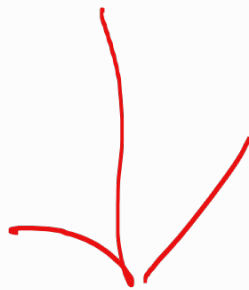
studentRepository.save(s);

—, — . findById(1);

—, —, findAll();

—, — . delete(s);

Difference between
these 3 are



JPA	Hibernate	SDJ
specification	ORM Framework	Spring module
No imple	Provides Imple	No, uses JPA only
No SQL generation	Generates SQL	Through Hibernate
No DB comm	Yes	— 1 —
can work alone	Can work alone	Needs JPA + an Imple
Not part of spring	Not part of spring	Yes It is